

Assignment – 1

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Intro to Web Development

Q.1-> Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Answer I =>

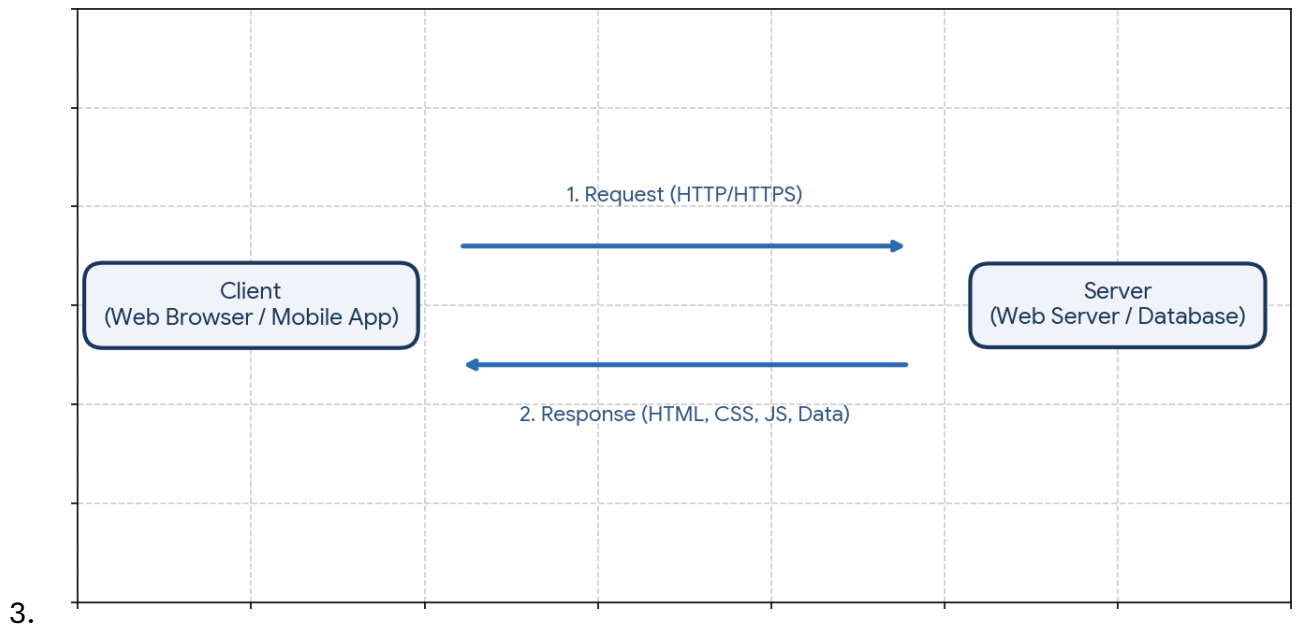
Key Differences Between Frontend, Backend, and Full Stack Development			
Parameters	Frontend Development	Backend Development	Full Stack Development
 Focus Area	• UI/UX elements	• Manage app logic, data, and security	• End-to-end software development
 Project Suitability	• Best for interactive, user-driven applications	• Ideal for data-intensive, complex software systems	• Suitable for MVPs, large-scale apps, & cost-effective software development
 Development Complexity	• Moderate—requires creative UI/UX expertise	• High—requires deep technical knowledge of databases & security protocols	• Higher—requires proficiency in both frontend and backend development
 Real-World Use Cases	• eCommerce websites, SaaS dashboards, and mobile apps	• Banking systems, cloud storage platforms, and APIs	• Web apps, mobile apps, and enterprise-grade software solutions
 Development Cost	Depends on business requirements, features and functionalities to be integrated, duration, location, and other project specifications.		

2.

Q.2-> Create a simple diagram showing how the client-server model works in web architecture.

Ans=>

Web Architecture Client-Server Model



Q.3-> Describe how a browser requests and displays a web page from a web server.

Ans-> Web browsers act as clients that request resources via network protocols, using a two-part process involving a Request Phase and a Rendering Phase to display web pages. The browser resolves domain names via DNS, establishes a TCP connection, and sends HTTP requests, subsequently parsing HTML and CSS to create DOM/CSSOM trees for visual display.

Q.4-> Identify and list the tools required to set up a web development environment. Explain the purpose of each.

- Ans=> **Back-End Frameworks**
- User Authentication
- Database Interactions
- **Text Editors**
- Syntax Highlighting
- Code Completion
- **Task Runners**
- Task Automation
- Configuration Files
- **Performance Optimization**
- Loading Times
- User Experience

- **Front-End Frameworks**
- UI Components
- Code Structuring
- **Content Management Systems**
- Content Creation
- No Code Required
- **Testing and Debugging**
- Automated Tests
- Error Identification
- **Integrated Development Environment**
- Code Editing
- Project Management
- **Version Control Systems**
- Collaboration
- Version History

1. Q.5-> Explain what a web server is and give examples of commonly used servers.

Ans=> A web server is a computer system or software that stores website files (like HTML, images, and CSS) and delivers them to user browsers over the internet via HTTP or HTTPS.

Apache Web Server

2. **IIS Web Server**
3. **Nginx Web Server**
4. **LiteSpeed Web Server**
5. **Apache Tomcat**
6. **Node.js**
7. **Lighttpd**
8. **OpenLiteSpeed**
9. **Jigsaw Server**
10. **Sun Java System Web Server**

Q-6-> Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans=> **Frontend Developer**

- **User Interface:** Builds layouts, buttons, and text using HTML, CSS, and JavaScript.
- **Client Interaction:** Makes sure the app works smoothly on all phone and computer screens.

- **API Connection:** Connects the visual screens to the server data using APIs.

Backend Developer

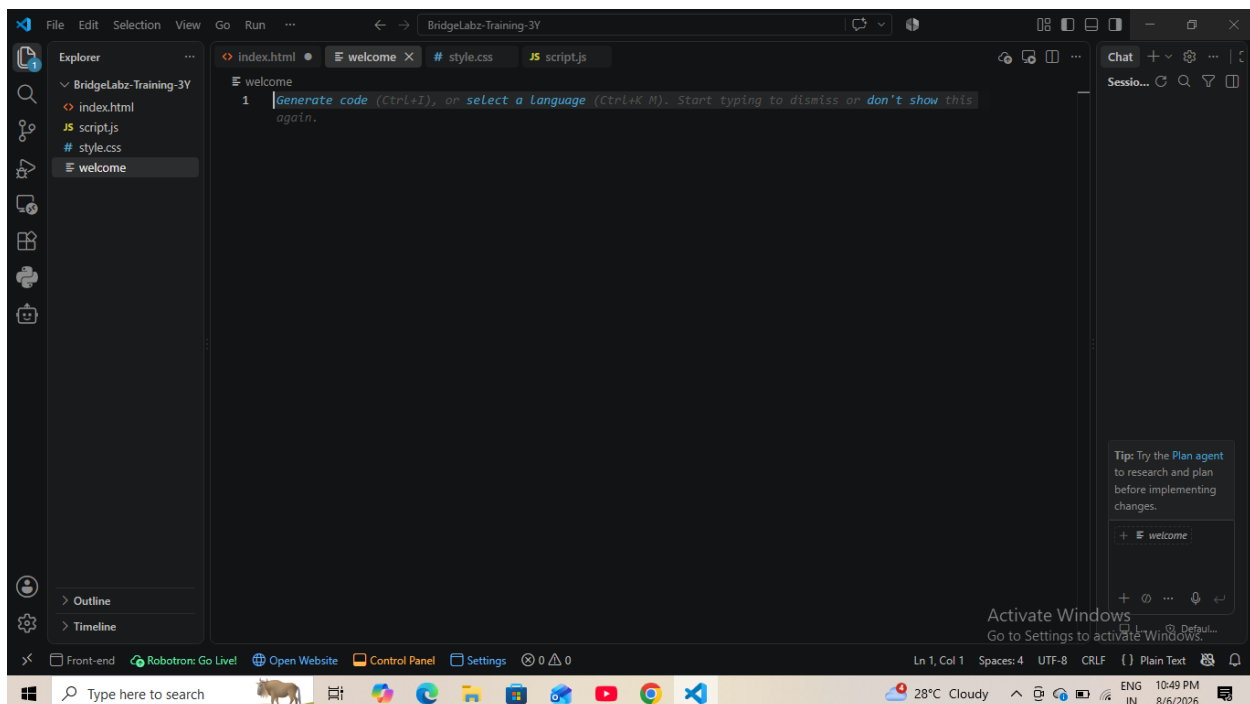
- **Server Logic:** Writes the core rules and operations that run on the server.
- **API Creation:** Builds endpoints so the frontend can request and send data safely.
- **App Security:** Adds login checks, data encryption, and defenses against web threats.

Database Administrator

- **Data Storage:** Sets up and maintains databases like SQL or MongoDB.
- **Performance Tuning:** Speeds up slow data queries and keeps systems fast.
- **Backup & Safety:** Protects data from loss and controls who can view or edit records.

Q.7-> Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Ans=>



Q.8-> Explain the difference between static and dynamic websites. Provide an example of

Each.

Ans=>

STATIC WEBSITE VERSUS DYNAMIC WEBSITE	
STATIC WEBSITE	DYNAMIC WEBSITE
A website whose web pages are coded in HTML and the content of each page is fixed and does not change unless it is edited and republished	A website whose web pages are generated in real time
Developed using client-side technologies such as HTML and CSS	Developed using client-side technologies as well as server-side scripting languages
Content remains unchanged unless it is changed from the source code	Content changes according to the client requests
Simple and easier to program	More complex and difficult to program
Does not allow many user interactions	Allow more user interaction
Do not access databases	Access information from a database
Cheaper to host	Costly to host
Difficult to update	Easier to update
Used for small-scale websites that do not require continuous changes	Suitable for large-scale e-commerce and social media websites

Q.9-> Research and list five web browsers. Explain how rendering engines differ between

- Them.

Ans=>

Blink: Google's open-source engine used in Chrome, Opera, Brave, Vivaldi, and other Chromium-based browsers. Known for speed, performance, and rapid adoption of modern web technologies.

- **Gecko:** Mozilla's engine powering Firefox, valued for strong standards compliance and consistent cross-platform rendering.
- **WebKit:** Apple's engine used in Safari and all iOS browsers, optimized for efficiency and smooth performance on Apple devices.
- **EdgeHTML:** Used in old versions of Microsoft Edge, designed to balance legacy compatibility with modern standards.
- **Other Engines:** Legacy or specialized engines like Presto(Opera Mini), Goanna(Vivaldi), and WebKit-based forks support specific platforms and add diversity to the web ecosystem.

Q.10-> Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans=>

